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$S = "()"$

Isko valid parentheses banana hai to
ya to hm ek '(' opening bracket ya
) closing bracket laga rhte hai.

$\therefore S = (())$

ek '(' opening bracket = valid
parentheses.

Output = 1

ex:-

$S = "((($

$\hookrightarrow ((()))$

three closing bracket
lagene honge.

Output = 3

20. $S = "()"$

$\nearrow \nearrow \nearrow$
 $\swarrow \swarrow \swarrow$

$() \rightarrow$ toh stack me pop
kar denge

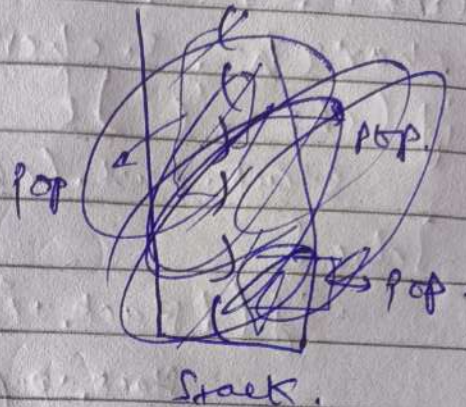
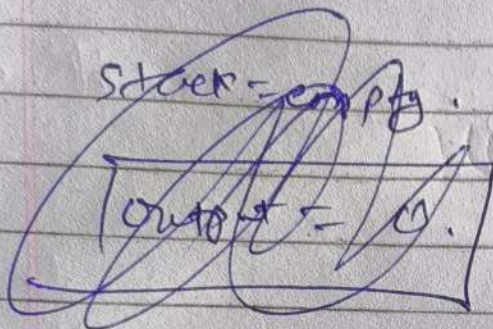
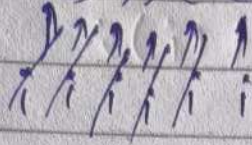
$\begin{array}{|c|} \hline) \\ \hline) \\ \hline (\\ \hline \end{array}$ pop.
stack.

end me jo bracket bachega wo
count hoga.

Output = 1

ex)

S = "()) (("



Output = 4.

Here

S = "()) (("

↳ pair one hai
pop ho jayega.

S = "()) (("

↳ pair nahi hai.

do '(' lagega. & do ')' lagega.

total 4 brackets chahiye.

~~Time~~ Time complexity = $O(n)$.

Space complexity = $O(n)$

↳ Because hm har ek bracket ko store kar rhe hai.

→ Isko hm ~~$O(1)$~~ $O(1)$ me convert kar skte hai.

Stack code

```
int count = 0;
for(int i = 0; i < s.length(); i++) {
    char ch = s.charAt(i);
    if (ch == '(') {
        stack.push(ch);
    } else {
        if (!stack.isEmpty()) {
            stack.pop();
        } else {
            count++;
        }
    }
}

return count + stack.size();
}
```

// No stack code:-

```
int open = 0;
int count = 0;
for(int i = 0; i < s.length(); i++) {
    char ch = s.charAt(i);
    if (ch == '(') {
        open++;
    } else {
        if (open > 0) {
            open--;
        } else {
            count++;
        }
    }
}

return count + open;
}
```